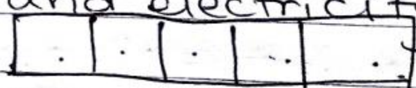
Inorganic Chemistry

Unit - 6
Transition Metals.

- d-block elements are called transition elements.
- Elements which involve the filling of electrons in d-orbital in penultimate shell i.e. shell before valence cell is called d-block elements.
- They exibitits transitional characteristics between ionic compounds forming s-block elements and covalent compound forming p-block elements.

General characteristics:-
- Hard, strong, High B.p and M.p
- conduct heat and electricity
- ductile



- Forms alloy with other metallic element.
- Electropositive enough to dissolve in mineral acids.
- Exibits catalytic properties.
- Form paramagnotic compounds as they have unpaired electrons.
- Most of them have great tendency to form complex ion.
- Variable oxidation states.
- Ions and compounds are coloured.
- General electronic configuaration:-
  $(n-1)\,d^{1-10}\,ns^{1-2}$

Oxidation state of Transition metals

- The number assigned to an element in a compound representing the number of electron lost or gained by an atom of the element of the compound is called oxidation state.
- The transition element show variable oxidation states bcz of following reasons:-
  - Valence electrons are present in two orbitals i.e. (n-1)d and ns
  - Energy difference between these two orbital is very less.
  - Electrons of both energy level can be used in bond formation.

Eg.

in copper, it's electronic configuration is [Ar] $3d^{10} 4s^1$.
It can lose 1 electron of 4s orbital easily and also can lose 1 electron of 3d orbital as energy required is very less. So it's oxidation states are +1 and +2.

- The oxidation state differs by unity.

Catalytic property of transition metals.

- Transition metal shows good catalytic properties because:-
  - Presence of vacant d-orbital
  - tendency to form complex compound
  - Ability to exibit variable valency

- Transition metal form unstable intermediate with their reactants as they have tendency to exibit variable valency and complex and the unstable intermediates provide alternate path having lower activation energy for reaction which increase rate of rxn.
- These intermediate decompose to give final product.
- Finely divided catalyst is used as they have large surface area.

Metal Complex Ions.

- Such Ion which has a metal ion in it's centre with a number of other molecules or ions surrounding it is called complex ion.
- Metal is known as central metal ion.
- The anions or molecules attached with central metal ion is known as ligands.
- They are bonded with co-ordinate covalent bond.
- The number of co-ordinate bonds being formed by metal ion at it's centre or no. of atoms directly attached to the central metal ion is called co-ordination number of metal complex ion.
- Transition metal are able to form a variety of complex ions. because:-

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(i) These ions are very small in size and have high charge density which helps in acceptance of lone pair of electrons.

(ii) They have vacant orbitals which accept lone pair donated by ligands

Ligands have active lone pair in outer energy level and they acts as lewis bases.

→ Complex Ion shape is the orientation of ligands or molecules that are attached to transition metals. and depends upon the co-ordination number and size of ligands.

→ The possible four shapes are:-

1. Linear

→ When two ligands and the metal ion are joined in a straight line.

→ Eg. When silver (II) is bonded with 2 ligands i.e. Ammonia, it forms tollens reagent

$[Ag(NH_3)_2]$

$$[H_3N - Ag - NH_3]^+$$

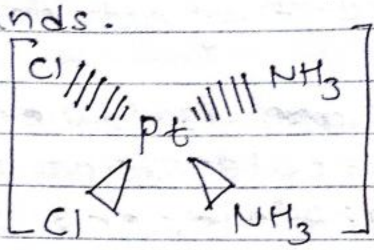
→ It's co-ordination number is 2.

2. Square planar shape

→ When central ion forms four bonds and ligands around the metal are all in same plane.

- It is completely flat
- It is obtained when ~~tec~~ ligands are attached with a d^8 electronic con~~o~~figuration metal.

Eg; Cisplatin :- It have platinum (II) as central metal ion and two ~~o~~ chlorine and two ammonia group on same side as ligands.



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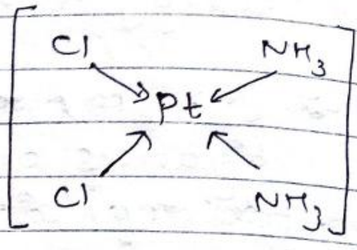
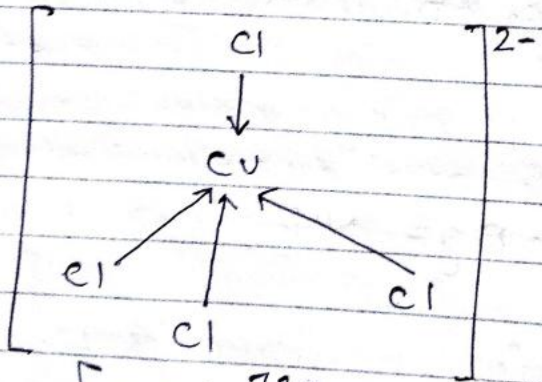


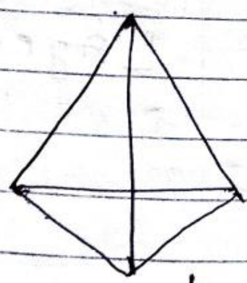
3. Tetrahedral shape.

- Metal ion is located at the central of four substituent atoms which forms corners of tetrahedron.
- found in complex of metal having d^0 or d^{10} electronic configuration.

Eg. $[CuCl_4]^{2-}$, $[CoCl_4]^{2-}$, etc.



$[CuCl_4]^{2-}$

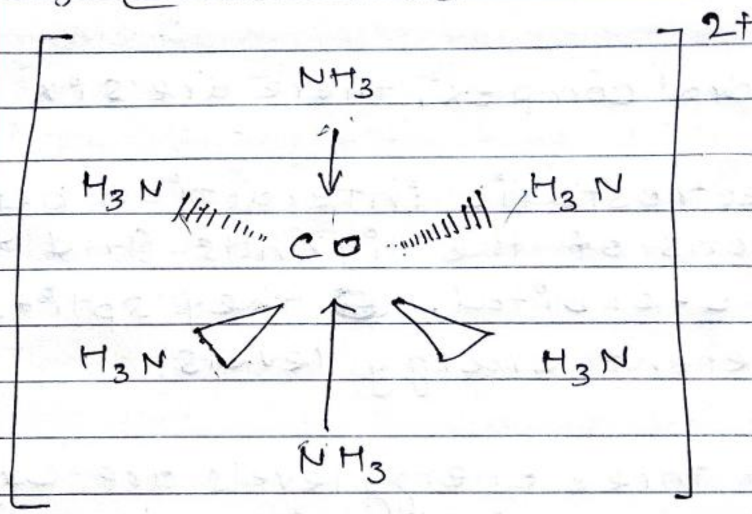


Tetrahedron

4. Octahedral Shape

- six ligands are symmetrically arranged around central metal.
- Forms shape of octahedron.
- Four of the ligands are in one plane with the fifth one above the plane and sixth one below the plane.

Eg. $[Co(NH_3)_6]^{2+}$



Crystal Field Theory

- It describes the electronic structure of transition metal complexes focusing on five valence d-orbitals.

It's main characteristics are

- It consider that metal-ligand bond is ionic. and electrostatic interaction occurs between them.

- There is repulsion between electron of d-orbital of metal and ligand's electron.

→ Negative end of dipole of neutral ligand is attracted toward metal atom. and behave ligand as point of -ve charge.

→ Arrangement of ligand around central metal atom is such a way that the repulsion should be minimum.

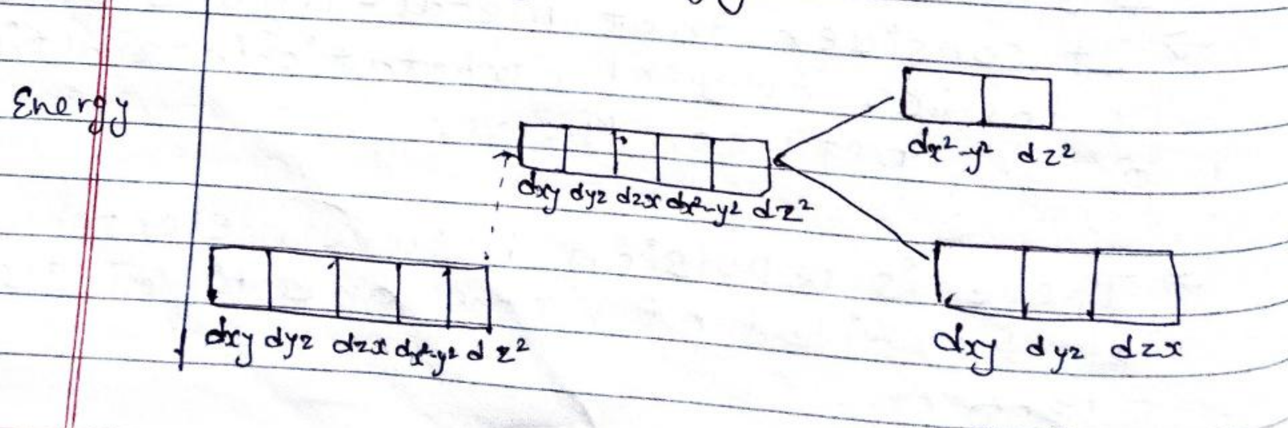
→ In octahedral complex, there are six ligands.

→ Due to electrostatic interaction between the electrons of the ligands and the lobes of d-orbital, they splits into two different energy levels.

→ The lower three energy levels are d_{xy}, d_{yz}, d_{xz} and collectively called t_{2g}.

→ The two upper energy levels are named as $d_{x^2-y^2}$ and d_{z^2} and collectively called e_g.

→ The difference in energy between the two set of d-orbitals is called crystal field splitting energy.



- This is used to explain many important properties of metal complexes like colours and magnetism.

Colour of transition metal compound.

- Most of the elements ions having unpaired electrons in their d-orbitals produce coloured compounds.
- Transition metal compounds are also coloured in both solid and aqueous state. due to following reasons:-

- In free state, they have five degenerate d-orbitals. Upon irradiation of light, the electron from lower set of d-orbital are excited to higher set of d-orbitals and energy difference between these two set of d-orbitals lies in visible range of spectrum.
- When one colour of visible (white) light is absorbed, another colour is reflected by them which is called complementary colour.